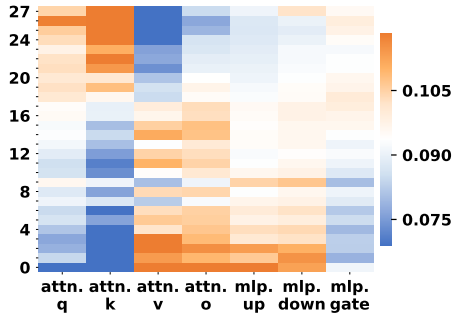
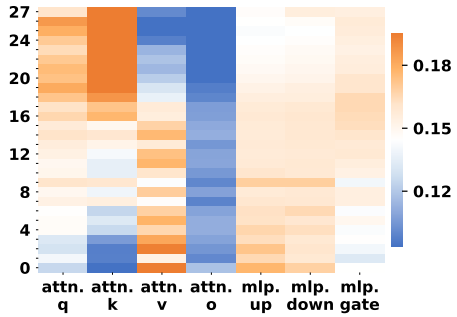


(a) Qwen2.5-7B-Instruct



(b) Distill-Qwen-7B

Figure 5: Parameter change rates of LLMs to the original ones across different modules. Darker orange colors indicate larger parameter changes.

improves performance on IFEval and CFBench, while our constraint-level variant achieves the best performance across all evaluated benchmarks.

Furthermore, our RM consistently outperforms LLM-as-a-Judge, demonstrating the superior effectiveness of our reward model. The constraint-level variant achieves substantial improvements over LLM-as-a-Judge on both IFEval and CFBench. Both instruction-level and constraint-level variants of our RM achieve competitive performance, with the constraint-level variant achieving the best overall results, indicating that our method is effective for different reward granularity. The superior performance of constraint-level rewards suggests that fine-grained constraint evaluation enables more precise optimization signals compared to instruction-level aggregation.

4.4.2 Generalization to Nested Structures

We conduct experiments to evaluate the performance of our method under nested logical-structure constraints. Although our training data only contains non-nested structures, LSRIF still improves performance on nested constraint structures: Selection_1 (depth 1), Selection_and_Chain_2 (depth 2), and Selection_and_Chain_3 (depth 3) from ComplexBench. As shown in Fig. 4, LSRIF maintains better performance across all depths compared to Base models. These results indicate that the improvements gained from training on non-nested structures generalize effectively to nested constraint structures, with the benefits becoming pronounced at higher nesting depths.

5 Interpretability Analysis

5.1 Parameter Change Patterns

Fig. 5 presents the relative parameter change rates across layers and modules after LSRIF training. The change rate is measured using the normalized Frobenius norm:

$$\Delta = \frac{\|W_{\text{after}} - W_{\text{before}}\|_F}{\|W_{\text{before}}\|_F} \times 100\%, \qquad (5)$$

where W_{before} and W_{after} denote the parameters before and after training. For a model with L layers, let $\Delta_m^{(l)}$ denote the change rate for module m at layer l.

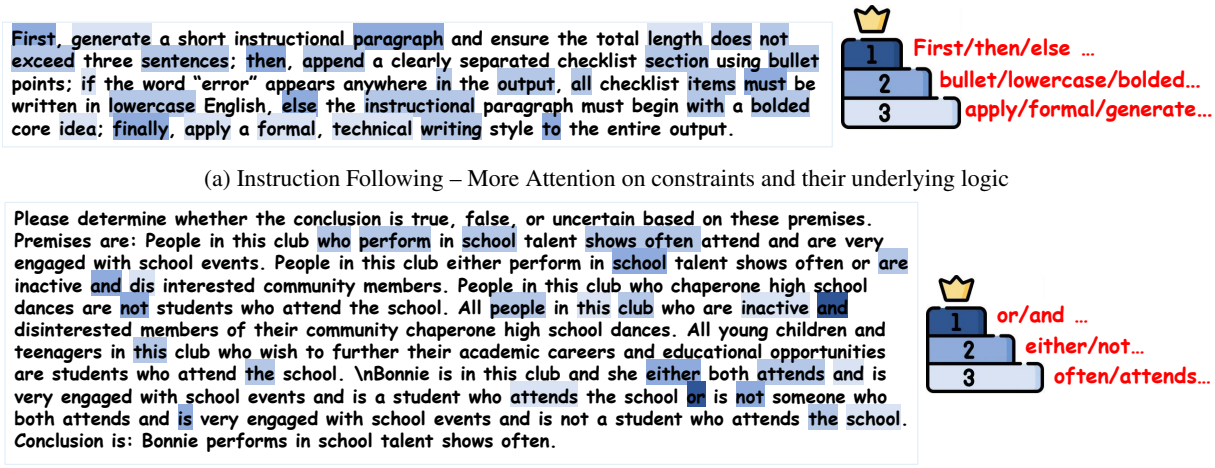
Attention vs. MLP Modules. A clear pattern observed in Fig. 5 is that attention modules undergo substantially larger parameter changes than MLP modules across most layers. In particular, the query and key projection matrices exhibit the highest change rates, while MLP up and down projections show comparatively smaller and more uniform updates:

$$\Delta_{\text{attn.q}}^{(l)}, \Delta_{\text{attn.k}}^{(l)} > \Delta_{\text{mlp.up}}^{(l)}, \Delta_{\text{mlp.down}}^{(l)}, \qquad (6)$$

Layer-wise Trends. This discrepancy between attention and MLP updates is consistent across layers. Although both module types display some variation along depth, attention-related parameters consistently dominate the overall magnitude of change, especially in lower and upper layers. In contrast, MLP parameters remain relatively stable throughout the network.

Model Consistency. The same trend holds for both Qwen2.5-7B-Instruct and Distill-Qwen-7B. While the distilled model shows larger absolute change magnitudes, the relative dominance of attention parameter updates over MLP updates remains consistent.

Overall, these results indicate that LSRIF primarily induces stronger updates in attention mechanisms, whereas MLP layers are affected to a much lesser extent.



(a) Instruction Following – More Attention on constraints and their underlying logic

(b) Logic Reasoning – More Attention on logical connectors

Figure 6: Comparison of attention importance changes for each token position in Qwen2.5-7B-Instruct before and after training on instruction following and logic reasoning tasks. Darker colors indicate greater increases.

5.2 Token-Level Information Flow Analysis

We analyze token-level information flow using gradient-based saliency attribution to quantify how training redirects attention to semantically critical tokens. For token x_i with embedding E_i, the attribution score is defined as

$$S_i = \left| \sum_{d=1}^{D} \frac{\partial L}{\partial E_{i,d}} \cdot E_{i,d} \right|. \tag{7}$$

The sequence-level loss function is defined as

$$L(x, y) = \sum_{t=1}^{|y|} \log P(y_t \mid y_{<t}, x). \tag{8}$$

The change in attention importance is measured as

$$\Delta S_i = S_i^{\text{after}} - S_i^{\text{before}}, \tag{9}$$

where higher values indicate greater increases in attention importance.

As shown in Fig. 6, training shifts attention from diffuse to concentrated patterns, directly corresponding to parameter changes in attention query and key modules (Fig. 5). For instruction following tasks, we observe a hierarchical attention increase across three token categories: logical connectors ("First", "then", "else") show the highest increase, constraint tokens ("bullet", "lowercase", "bolded") show moderate increase, and action verbs ("apply", "formal") show lower increase. For logic reasoning tasks, we observe a similar hierarchical pattern: logical operators ("or", "and") show the highest increase, followed by choice/negation terms ("either", "not") and descriptive predicates ("attends").

This hierarchical pattern indicates that the model prioritizes structural elements encoding logical relationships, aligning with structure-aware reward modeling. The substantial updates to $\Delta^{(l)}_{\text{attn.q}}$ and $\Delta^{(l)}_{\text{attn.k}}$ enable query and key representations that prioritize tokens encoding logical structures. The attention mechanism computes query-key similarities where query and key projections are updated to maximize attention weights for structural tokens, validating that LSRIF adapts attention mechanisms to capture constraint relationships rather than merely adjusting output representations.

6 Conclusion

In this work, we propose LSRIF, a logic-structured training framework. We construct LSRINSTRUCT, a multi-constraint instruction dataset covering parallel, sequential, and conditional constraint logic structures, and design LSRM, structure-aware reward modeling that aligns training signals with logical execution semantics. LSRIF improves instruction following in both in-domain and out-of-domain settings, while also enhancing general reasoning ability. We also conduct attention and token-level interpretability analysis for model performance improvements.

7 Limitations

Our study has following main limitations. First, due to computational constraints, we do not evaluate our method on larger models such as 70B+, and validation at this scale would further strengthen the credibility and robustness of our approach. Sec-

ond, our training data is primarily English. While results on CFBench indicate that logic-structured training can generalize to other languages, we encourage the community to construct multilingual logic-structured instruction datasets to more systematically assess and extend cross-lingual generalization.

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